

SIMPLE USER MANUAL - HOW TO OPERATE THE SOFTWARE

This short guide shows how to log in, configure the demo, and run the main operating scenarios.

1. Start and log in

- 1.1 Open the application in a browser (local PC or Streamlit Community Cloud URL).
- 1.2 On the login screen, enter Username: pocuser
- 1.3 Enter Password: poc123
- 1.4 Click Login.
- 1.5 Optional: use Light / Dark at the top right to match the room lighting.
- 1.6 After login you should see the title Cognitive Radio Channel Selection under Jamming.

2. Understand the screen layout

- 2.1 Left sidebar = user, display view, configuration, and run controls.
- 2.2 Main area (Architecture view) shows three nodes: Jammer, Receiver, Transmitter.
- 2.3 Jammer (red) = adversary ECM threat. Choose profile and power, then Emit Jam or Silence Jam.
- 2.4 Receiver (blue) = sensing node. Shows FREE / DEGRADED / BLOCKED counts and the channel spectrum map.
- 2.5 Transmitter (green) = ECCM actuator. Shows active carrier, TX power, hopping, and neural / hybrid decision.
- 2.6 Colour meaning on the map: green = FREE, amber = DEGRADED, red = BLOCKED.

3. Recommended starting configuration

- 3.1 In Display View, select System Architecture & Node View.
- 3.2 Set Number of channels (N) to 8.
- 3.3 Set Decision engine to HYBRID.
- 3.4 Leave Cognitive / adaptive hopping OFF for the first scenarios (turn ON later in Scenario 8).
- 3.5 Leave SINR FREE threshold at about 10 dB and SINR BLOCKED threshold at about 3 dB.
- 3.6 Leave Maximum TX power at 30 dB unless you are demonstrating power limits.
- 3.7 Click Apply configuration and wait if ML training is shown.
- 3.8 Important rule: after changing N, engine, hopping, or thresholds, always click Apply configuration.

4. Scenario - Clean spectrum (no jamming)

- 4.1 Purpose: show a quiet spectrum before any ECM threat.
- 4.2 In the Jammer node, click Silence Jam.
- 4.3 In the sidebar under Run control, click Single step once or twice.
- 4.4 Look at the Receiver channel spectrum map.
- 4.5 Expected result: most or all channels are FREE (green). Transmitter shows an active carrier on a usable channel.

5. Scenario - Spot jam on one channel

- 5.1 Purpose: show the radio leaving one jammed channel.
- 5.2 In the Jammer node, set ECM Threat Profile to SPOT.
- 5.3 If target-channel controls are shown, select one channel (for example the current active carrier).
- 5.4 Set Jam power (linear a.u.) to about 40.
- 5.5 Click Emit Jam.

- 5.6 If needed, click Single step once more so the map refreshes.
- 5.7 Expected result: the jammed channel turns BLOCKED (red). Cognitive Radio recommends another FREE channel. Transmitter retunes away from the jammed channel.

6. Scenario - Multi-spot jam on several channels

- 6.1 Purpose: show avoidance when more than one channel is attacked.
- 6.2 Set ECM Threat Profile to MULTI_SPOT.
- 6.3 Select two or more target channels (for example CH0 and CH1).
- 6.4 Click Emit Jam, then Single step if needed.
- 6.5 Expected result: selected channels are BLOCKED. Receiver free-link count drops. Transmitter moves to a remaining FREE or usable channel.

7. Scenario - Sweep jam over time

- 7.1 Purpose: show the cognitive loop adapting as the jam moves.
- 7.2 Set ECM Threat Profile to SWEEP.
- 7.3 Click Emit Jam.
- 7.4 In Run control, set Auto steps to 5 or 10.
- 7.5 Click Run auto steps and watch the Receiver map.
- 7.6 Expected result: the BLOCKED channel moves over time. The Cognitive Radio keeps recommending a usable channel as conditions change.

8. Scenario - Barrage jam and increase power

- 8.1 Purpose: show the all-channels-stressed path and power advice.
- 8.2 Set ECM Threat Profile to BARRAGE.
- 8.3 Set Jam power high (for example 40).
- 8.4 Click Emit Jam.
- 8.5 Click Single step several times.
- 8.6 Watch Transmitter TX power and the neural / hybrid decision card.
- 8.7 Expected result: many or all channels are BLOCKED. Decision often becomes INCREASE_POWER while headroom remains, and TX power rises.
- 8.8 If TX power is already at Maximum TX power, expected status is ALL_BLOCKED.

9. Scenario - Cognitive / adaptive hopping

- 9.1 Purpose: show hop-set adaptation under changing interference.
- 9.2 In Configuration, turn Cognitive / adaptive hopping ON.
- 9.3 Click Apply configuration and wait for ML training if shown.
- 9.4 Emit a SWEEP or MULTI_SPOT jam.
- 9.5 Run several auto steps.
- 9.6 Look at Transmitter Hopping Mode and Active Hop Pool.
- 9.7 Expected result: hop pool is shown; jammed members can be dropped and cleaner channels added. Transmitter follows the hop set.

10. Scenario - Compare RULES, ML, and HYBRID

- 10.1 Purpose: show the same operator interface with different decision engines.
- 10.2 Set Decision engine to RULES, click Apply configuration, emit a MULTI_SPOT jam, then Single step.
- 10.3 Note the status / action and recommended channel.
- 10.4 Change Decision engine to ML, Apply configuration, repeat the same jam and step.
- 10.5 Change Decision engine to HYBRID and repeat once more.
- 10.6 Expected result: response style stays inspectable (status, message, channel / power advice). HYBRID

uses ML with a safety gate.

11. Scenario - Switch to Operator Spectrum View

- 11.1** Purpose: use the classic spectrum map and measurement table.
- 11.2** In Display View, select Operator Spectrum View.
- 11.3** Use sidebar Jam injection (Apply jam / Clear jam) in this view.
- 11.4** Click Single step or Run auto steps as needed.
- 11.5** Expected result: channel cards, metrics, and decision history update in the spectrum-oriented layout.

12. Scenario - Reset and log out

- 12.1** To clear the threat: click Silence Jam (Architecture view) or Clear jam (Spectrum view).
- 12.2** To restart cleanly: click Reset simulation, then wait for retraining if prompted.
- 12.3** To leave the session: click Logout in the sidebar.
- 12.4** Expected result: simulation returns to a clean start, or the login screen appears after logout.

13. How to read common status codes

- 13.1** OK - at least one FREE channel is recommended.
- 13.2** DEGRADED_OK - no FREE channel; a usable DEGRADED channel is recommended.
- 13.3** INCREASE_POWER - all blocked at current power; raise TX power.
- 13.4** ALL_BLOCKED - still blocked at maximum allowed power.
- 13.5** HOLD - keep the current allocation (if shown).

14. Short presenter checklist

- 14.1** Log in -> show clean spectrum (Silence Jam -> Single step).
- 14.2** Spot or Multi-spot Emit Jam -> show red BLOCKED tiles -> show Transmitter retune.
- 14.3** Sweep + auto steps -> show blocked cell moving.
- 14.4** Barrage -> show INCREASE_POWER and rising TX power.
- 14.5** Optional: enable hopping; optional: compare RULES vs HYBRID.
- 14.6** Remind the audience: this is a software-only demonstration; no radiated RF.

15. Quick troubleshooting

- 15.1** Map does not change after Emit Jam -> click Single step.
- 15.2** Settings seem ignored -> click Apply configuration after changing them.
- 15.3** ML not trained message -> Apply configuration or Reset simulation.
- 15.4** Need a clean demo again -> Silence / Clear jam, then Reset simulation.
- 15.5** Local start command (if running on a PC): `python -m streamlit run app.py` then open `http://localhost:8501`